

Dear Dr. Wake,

Please consider this manuscript “Phenological sensitivity to climate change disrupts species coexistence” as as Perspective in *Nature Climate Change*.

Shifts in phenology—the seasonal timing of organisms’ life cycle events—are one of the most widespread biological indicators of climate change, yet the strength of these shifts vary substantially among species. These phenological differences among taxa have often been overlooked in ecological theory, but growing evidence suggests that they may play a major role in dictating community interactions and shaping the structure and function of ecological communities (1; 2; 3). While this recognition has spurred advances towards integrating phenology into community assembly models, current frameworks are not designed to capture the dynamic interactions between the physical environment, phenology and competition, and therefore struggle to make inference on how species interactions will shift with climate change.

Here, we integrate empirical data on seed germination phenology with a novel extension of a well established community assembly model to develop a simple and flexible framework to fill this gap. We demonstrate how modeling **differences in species’ phenological sensitivity** (i.e., their phenological response or reaction norm to environmental variation) in place of the expressed phenological differences themselves can overcome current limitations in community assembly theory. With this subtle, yet essential, adjustment we show that any phenologically-mediated patterns of coexistence under historical climate conditions will degrade as climate warms while providing a biologically realistic mechanism to project changes in these species iterations into the future.

Importantly, species-level estimates phenological sensitivity can be readily estimated with simple experiments, and our model can be leveraged to incorporate multiple, interacting environmental cues, variable environmental stressors, and more complex species interactions to develop predictive frameworks that can ultimately be effectively used to inform species conservation and ecological management. Our manuscript details a path forward to do this effectively and advance ecological theory, climate change biology, and the translation ecology that links them together.

We have previous discussed this manuscript with *Nature Climate Change* Editor Dr. Tegan Armarego-Marriott during the 2024 meeting of Ecological Society of America meeting in Long Beach, CA, and feel it would be of broad interest to the readership of *Nature Climate Change*. The main text of this manuscript is 2001 words in length and it contains two figures. It is co-authored by M.J. Donahue and E.M. Wolkovich and is not under consideration elsewhere. We hope that you will find it suitable for publication in *Nature Climate Change*, and look forward to hearing from you.

Sincerely,

Daniel Buonaiuto

References

- [1] Cleland, E. E. and Wolkovich, E. Effects of phenology on plant community assembly and structure. *Annual Review of Ecology, Evolution, and Systematics* **55**(Volume 55, 2024), 471–492 (2024).
- [2] Blackford, C., Germain, R. M., and Gilbert, B. Species differences in phenology shape coexistence. *The American Naturalist* **195**(6), E168–E180 (2020). PMID: 32469665.
- [3] Rudolf, V. H. W. The role of seasonal timing and phenological shifts for species coexistence. *Ecology Letters* (**2019**).